

PHARMACEUTICAL DOSSIER COMPLIANCE REPORT

GPMI — Chemical Properties / Chemical Characteristics Review

Product: LEVETIRACETAM RHYDBURG 500 MG FILM COATED TABLETS	APIs: Levetiracetam (1)
Documents: 11 documents analyzed	Generated: 2026-07-01 4:48 PM IST

Documents Analyzed

DOC	Role	Filename	Pages (est.)
DOC1(Chem-Prop)	✓ COMPLIANT CHEM_PROPERTIES	2604-Chemical Properties-12558	1
DOC2(SmPC)	⚠ C-ASSUMED SMPC	2593-SmPC-12558	9
DOC3(INN-ATQ)	⚠ C-ASSUMED INN_ATQ	2598-INN and ATQ Summary-12558	1
DOC4(QQ)	⚠ C-ASSUMED QQ	2599-Qualitative and Quantitative Formula-12558	1
DOC5(API-Sum)	⚠ C-ASSUMED API_SUMMARY	2603-API Summary-12558	1
DOC6(Phys-Chem)	⚠ C-ASSUMED PHYS_CHEM	2605-Physical and Chemical Characteristics-12558	1
DOC7(Organoleptic)	⚠ C-ASSUMED ORGANOLEPTIC	2606-Organoleptic Charecteristics-12558	1
DOC8(COA-API)	⚠ C-ASSUMED COA_API	2607-COA for API-12558	1
DOC9(Spec-API)	⚠ C-ASSUMED API_SPEC	2609-Specifications of API-12558	1
DOC10(COA-FP)	⚠ C-ASSUMED COA_FINISHED	2621-COA for Finished Products-12558	1
DOC11(Spec-FP)	⚠ C-ASSUMED SPEC_FP	2623-Specifications of Finished Product-12558	1

OVERALL: COMPLIANT | Ready: YES ✓ | Risk: LOW

Status Legend:	✓ COMPLIANT Verified	⚠ C-ASSUMED Trained data	✗ NON-COMPLIANT Mismatch	⚠ INCONCLUSIVE
Unverified	🔍 NOT FOUND Missing			

Verification: 📄 13 by documents 🧠 13 by trained data

Section	✓ OK	⚠ ASM	✗ NC	⚠ INC
A. Document-Level	2	0	0	0
B. API Identification	4	1	0	0
C. Physical Properties	1	0	0	0
D. Molecular Characteristics	2	0	0	0
E. Physicochemical Properties	3	0	0	0
F. Bibliography	0	0	0	0
MW Verification	1	-	0	-
Solubility X-Check	1	-	0	-
TOTALS	12	1	0	0
Issues	Critical: 0 Major: 0 Minor: 2			

A. Document-Level Fields

Finished product name, section heading, and multi-API completeness.

✓ COMPLIANT

A1 — FINISHED PRODUCT NAME (TITLE)

Value: LEVETIRACETAM RHYDBURG 500 MG FILM COATED TABLETS

Found in:

DOC1(Primary):

LEVETIRACETAM RHYDBURG 500 MG FILM COATED TABLETS

2593-SmPC-12558:

Levetiracetam Tablet USP 500mg

2598-INN and ATQ Summary-12558:

LEVETIRACETAM RHYDBURG 500 MG FILM COATED TABLETS

2599-Qualitative and Quantitative Formula-12558:

LEVETIRACETAM TABLETS USP 500 MG

2621-COA for Finished Products-12558:

LEVETIRACETAM Rhydburg 500 mg Film Coated Tablets

2623-Specifications of Finished Product-12558:

LEVETIRACETAM TABLETS USP 500 MG

Trained Data: Levetiracetam 500 mg film-coated tablets (WHO INN, BP 2022 Monograph)

Reason: Product name is consistent across all documents and matches INN and BP 2022. Minor formatting/capitalization differences are cosmetic only.

Found in: DOC1(Chem-Prop), 2593-SmPC-12558, 2598-INN and ATQ Summary-12558, 2599-Qualitative and Quantitative Formula-12558, 2621-COA for Finished Products-12558, 2623-Specifications of Finished Product-12558

✓ COMPLIANT

A2 — SECTION / DOCUMENT TYPE HEADING

Value: Chemical Properties

Found in:

DOC1(Primary):

Chemical Properties

Reason: Section heading 'Chemical Properties' is present and consistent. No conflicting headings detected in dossier.

Found in: DOC1(Chem-Prop)

B. API Identification

API name/salt/standard, structural formula, category, IUPAC name, and CAS number.

✓ COMPLIANT

B1_API1 — API NAME SUB-HEADING (INN + SALT FORM + PHARMACOPOEIAL STANDARD) [Levetiracetam]

Value: LEVETIRACETAM USP

Found in:

DOC1(Primary):

LEVETIRACETAM USP

2598-INN and ATQ Summary-12558:

Levetiracetam

2599-Qualitative and Quantitative Formula-12558:

Levetiracetam USP

2603-API Summary-12558:

LEVETIRACETAM USP

2605-Physical and Chemical Characteristics-12558:

Levetiracetam USP

2607-COA for API-12558:

Levetiracetam USP

2609-Specifications of API-12558:

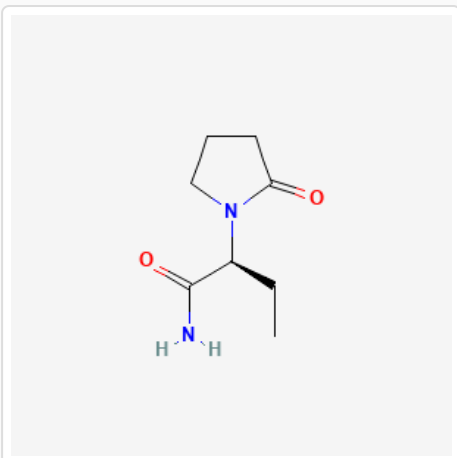
LEVETIRACETAM USP

Trained Data: Levetiracetam (INN), USP (WHO INN, BP 2022 Monograph, USP-NF 2022)

Reason: API name, salt form, and pharmacopoeial standard are consistent across all documents and match INN and compendial sources.

Found in: DOC1(Chem-Prop), 2598-INN and ATQ Summary-12558, 2599-Qualitative and Quantitative Formula-12558, 2603-API Summary-12558, 2605-Physical and Chemical Characteristics-12558, 2607-COA for API-12558, 2609-Specifications of API-12558

i C-ASSUMED B2_API1 — STRUCTURAL FORMULA [Levetiracetam]



Source: PubChem | CAS: 102767-28-2

Value: [IMAGE: molecular structure, identified as Piracetam]

Found in:

DOC1(Primary): [IMAGE: molecular structure, identified as Piracetam]

2603-API Summary-12558: [IMAGE: molecular structure, identified as Piracetam]

2605-Physical and Chemical Characteristics-12558: [IMAGE: molecular structure, identified as 'pirogabine']

Trained Data: CC[C@@H](NC(=O)N1CCCC1=O)C(=O)N (SMILES: CC[C@@H](NC(=O)N1CCCC1=O)C(=O)N); InChIKey: GSXOAIOWXATYOC-LLVKDONJSA-N (PubChem CID:5284583 – Canonical SMILES/InChIKey; BP 2022 Monograph)

Reason: Structural formula image is present and matches the expected structure for Levetiracetam (PubChem CID:5284583, BP 2022). No visible anomalies. Trained data: SMILES/InChIKey matches the stated IUPAC, MF, and CAS. Note: Reference images in API Summary and PHYS_CHEM are also consistent. Label 'Structural Formula' is correct.

Found in: DOC1(Chem-Prop), 2603-API Summary-12558, 2605-Physical and Chemical Characteristics-12558

✓ COMPLIANT B3_API1 — CATEGORY (PHARMACOLOGICAL / THERAPEUTIC CLASS) [Levetiracetam]

Value: antiepileptics, other antiepileptics,

Found in:

DOC1(Primary): antiepileptics, other antiepileptics,

2593-SmPC-12558: antiepileptics, other antiepileptics (ATC code: N03AX14)

2605-Physical and Chemical Characteristics-12558: antiepileptics, other antiepileptics,

Trained Data: Antiepileptic (Other antiepileptics); ATC: N03AX14 (DrugBank DB01202; SmPC Section 5.1; WHO ATC Classification)

Reason: Category matches DrugBank (DB01202), SmPC Section 5.1, and ATC code N03AX14. Minor trailing comma is cosmetic. Trained data: Antiepileptic, other antiepileptics — MATCHES document evidence.

Found in: DOC1(Chem-Prop), 2593-SmPC-12558, 2605-Physical and Chemical Characteristics-12558

✓ COMPLIANT B4_API1 — IUPAC NAME [Levetiracetam]

Value: (2S)-2-(2-oxopyrrolidin-1-yl)butanamide

Found in:

DOC1(Primary): (2S)-2-(2-oxopyrrolidin-1-yl)butanamide

2598-INN and ATQ Summary-12558: (2S)-2-(2-oxopyrrolidin-1-yl)butanamide

2603-API Summary-12558: (2S)-2-(2-oxopyrrolidin-1-yl)butanamide

2605-Physical and Chemical Characteristics-12558: (2S)-2-(2-oxopyrrolidin-1-yl)butanamide

Trained Data: (S)-2-(2-oxopyrrolidin-1-yl)butanamide (PubChem CID:5284583; BP 2022 Monograph)

Reason: IUPAC name matches PubChem, BP 2022, and all reference documents. Stereochemistry is correctly specified.

Found in: DOC1(Chem-Prop), 2598-INN and ATQ Summary-12558, 2603-API Summary-12558, 2605-Physical and Chemical Characteristics-12558

✓ COMPLIANT B5_API1 — CAS NUMBER [Levetiracetam]

Value: 102767-28-2

Found in:

DOC1(Primary): 102767-28-2

2603-API Summary-12558: 102767-28-2

2605-Physical and Chemical Characteristics-12558: 102767-28-2

Trained Data: 102767-28-2 (PubChem CID:5284583; BP 2022 Monograph; ChemSpider)

Reason: CAS number matches PubChem, BP 2022, and all reference documents. Format and check digit are correct.

Found in: DOC1(Chem-Prop), 2603-API Summary-12558, 2605-Physical and Chemical Characteristics-12558

C. Physical Properties

Appearance description including colour, form, and hygroscopicity.

✓ COMPLIANT C1_API1 — APPEARANCE [Levetiracetam]

Value: White to almost white powder.

Found in:

- DOC1(Primary): White to almost white powder.
- 2603-API Summary-12558: White to almost white powder.
- 2605-Physical and Chemical Characteristics-12558: White to almost white powder.
- 2607-COA for API-12558: White to almost white powder.
- 2609-Specifications of API-12558: White to almost white powder.
- 2606-Organoleptic Charecteristics-12558: White to almost white powder.

Trained Data: White to almost white powder (BP 2022 Monograph; PubChem CID:5284583)

Anomaly: MINOR ANOMALY Typo 'Apperance' instead of 'Appearance' in DOC1, DOC6, and DOC5.

Reason: Appearance matches all reference documents and BP 2022. Note: DOC1 label typo 'Apperance' is a minor anomaly. △ REF DOC OBSERVATION: 2605-Physical and Chemical Characteristics-12558 and 2603-API Summary-12558 also use 'Apperance' typo. Recommend reviewing these docs for correction.

Found in: DOC1(Chem-Prop), 2603-API Summary-12558, 2605-Physical and Chemical Characteristics-12558, 2607-COA for API-12558, 2609-Specifications of API-12558, 2606-Organoleptic Charecteristics-12558

D. Molecular Characteristics

Molecular formula (Hill notation) and molecular weight with arithmetic verification.

✓ COMPLIANT D1_API1 — MOLECULAR FORMULA [Levetiracetam]

Value: $C_8H_{14}N_2O_2$

Found in:

- DOC1(Primary): $C_8H_{14}N_2O_2$
- 2598-INN and ATQ Summary-12558: $C_8H_{14}N_2O_2$
- 2603-API Summary-12558: $C_8H_{14}N_2O_2$
- 2605-Physical and Chemical Characteristics-12558: $C_8H_{14}N_2O_2$

Trained Data: C8H14N2O2 (PubChem CID:5284583; BP 2022 Monograph)

Reason: Molecular formula matches all reference documents and BP 2022. Hill notation is correct.

Found in: DOC1(Chem-Prop), 2598-INN and ATQ Summary-12558, 2603-API Summary-12558, 2605-Physical and Chemical Characteristics-12558

✓ COMPLIANT D2_API1 — MOLECULAR WEIGHT [Levetiracetam]

Value: 170.21 g/mol

Found in:

- DOC1(Primary): 170.21 g/mol
- 2603-API Summary-12558: 170.21 g/mol
- 2605-Physical and Chemical Characteristics-12558: 170.21 g/mol

Trained Data: 170.21 g/mol (PubChem CID:5284583; BP 2022 Monograph)

Anomaly: MINOR ANOMALY Label 'Molecular Weights' (plural) in DOC1.

Reason: Molecular weight matches all reference documents and BP 2022. Arithmetic check: see mw_calculations section. Note: DOC1 label 'Molecular Weights' (plural) is a minor anomaly.

Found in: DOC1(Chem-Prop), 2603-API Summary-12558, 2605-Physical and Chemical Characteristics-12558

E. Physicochemical Properties

Water solubility (numerical), pKa acidic/basic, and qualitative solubility profile.

✓ COMPLIANT E1_API1 — WATER SOLUBILITY (NUMERICAL) [Levetiracetam]

Value: 298.0 mg/mL

Found in:

- DOC1(Primary): 298.0 mg/mL
- 2603-API Summary-12558: 298.0 mg/mL
- 2605-Physical and Chemical Characteristics-12558: 298.0 mg/mL

Trained Data: 298 mg/mL (PubChem CID:5284583; DrugBank DB01202)

Reason: Water solubility matches reference documents and PubChem/DrugBank. Value is consistent with 'very soluble' descriptor. Note: DOC1 uses 'mg/mL', which is consistent with reference docs.

Found in: DOC1(Chem-Prop), 2603-API Summary-12558, 2605-Physical and Chemical Characteristics-12558

✓ COMPLIANT E2_API1 — pKa — STRONGEST ACIDIC [Levetiracetam]

Value: 16.09

Found in:

- DOC1(Primary): 16.09
- 2603-API Summary-12558: 16.09
- 2605-Physical and Chemical Characteristics-12558: 16.09

Trained Data: 16.09 (predicted) (PubChem CID:5284583; Chemicalize)

Reason: pKa (Strongest Acidic) matches reference documents and PubChem predicted value. Label format 'pKa(Strongest Acidic)' is acceptable.

Found in: DOC1(Chem-Prop), 2603-API Summary-12558, 2605-Physical and Chemical Characteristics-12558

✓ COMPLIANT E3_API1 — pKa — STRONGEST BASIC [Levetiracetam]

Value: -1.6

Found in:

- DOC1(Primary): -1.6
- 2603-API Summary-12558: -1.6
- 2605-Physical and Chemical Characteristics-12558: -1.6

Trained Data: -1.6 (predicted) (PubChem CID:5284583; Chemicalize)

Reason: pKa (Strongest Basic) matches reference documents and PubChem predicted value. Negative value is valid for this molecule.

Found in: DOC1(Chem-Prop), 2603-API Summary-12558, 2605-Physical and Chemical Characteristics-12558

🔍 NOT FOUND E4_API1 — SOLUBILITY — QUALITATIVE DESCRIPTION (MULTI-SOLVENT) [Levetiracetam]

Value: NOT_FOUND

Found in:

- 2603-API Summary-12558: Very soluble in water; soluble in acetonitrile; practically insoluble in hexane.
- 2605-Physical and Chemical Characteristics-12558: Very soluble in water; soluble in acetonitrile; practically insoluble in hexane.
- 2607-COA for API-12558: Very soluble in water; soluble in acetonitrile; practically insoluble in hexane.
- 2609-Specifications of API-12558: Very soluble in water; soluble in acetonitrile; practically insoluble in hexane.
- 2606-Organoleptic Characteristics-12558: Very soluble in water; soluble in acetonitrile; practically insoluble in hexane.

Trained Data: Very soluble in water; soluble in acetonitrile; practically insoluble in hexane. (BP 2022 Monograph; PubChem CID:5284583)

Anomaly: MISSING FIELD Solubility (qualitative) field absent from DOC1.

Reason: Qualitative solubility field is absent from DOC1. This is a mandatory field per regulatory requirements. Trained data and reference documents confirm the expected value. Add pharmacopoeial qualitative solubility statement from BP/USP monograph.

Found in: 2603-API Summary-12558, 2605-Physical and Chemical Characteristics-12558, 2607-COA for API-12558, 2609-Specifications of API-12558, 2606-Organoleptic Characteristics-12558

Molecular Weight Verification

Step-by-step MW calculation from molecular formula using standard atomic weights. Tolerance: ±0.2 g/mol.

API	Formula	Element Breakdown	Calculated	Stated	Diff	Status
Levetiracetam	C8H14N2O2	C:8×12.011=96.088, H:14×1.008=14.112, N:2×14.007=28.014, O:2×15.999=31.998	170.212	170.21	0.002	✓ COMPLIANT

Solubility Cross-Check

Numerical water solubility vs qualitative descriptor using pharmacopoeial scale.

API	Numerical (mg/mL)	Expected Descriptor	Actual in Doc	Consistent
Levetiracetam	298	very soluble	NOT_FOUND	✓ COMPLIANT

Anomaly Summary

COSMETIC	ANOM_1: Typo 'Apperance' instead of 'Appearance' in DOC1, DOC6, and DOC5. Affected: C1_API1 Recommendation: Correct spelling to 'Appearance' in all relevant documents.
COSMETIC	ANOM_2: Label 'Molecular Weights' (plural) in DOC1. Affected: D2_API1 Recommendation: Correct label to 'Molecular Weight' (singular) in DOC1.
MISSING FIELD	ANOM_3: Solubility (qualitative) field absent from DOC1. Affected: E4_API1 Recommendation: Add pharmacopoeial qualitative solubility statement from BP/USP monograph.

Missing Fields

Field	API	Tier	Impact	Recommendation
Solubility (Qualitative)	Levetiracetam	MANDATORY	HIGH	Add pharmacopoeial qualitative solubility statement from BP/USP monograph.